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Intelligent Video Analytics Platform - Week 3

FELLOW	TRACK & ASSIGNMENT	PROJECT STATUS
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1. INTRODUCTION

Multi-object tracking (MOT) is the task of locating multiple objects in a video and assigning each one a consistent identity across frames. The dominant modern approach is tracking-by-detection: an object detector (e.g. YOLO) finds objects in each frame independently, and a separate tracking algorithm links those per-frame detections into continuous trajectories over time. This report discusses four influential tracking algorithms that represent this approach's evolution — SORT, DeepSORT, ByteTrack, and BoT-SORT — and compares them on accuracy, speed, and typical applications.

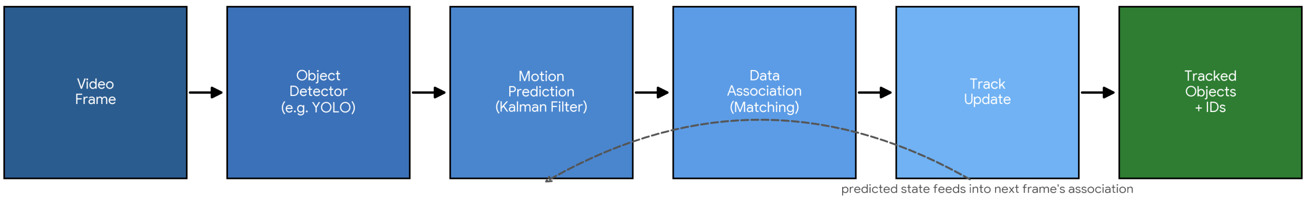

Figure 1 — The generic tracking-by-detection pipeline all four algorithms build on. They differ mainly in how the Data Association step is done.

2. SORT (SIMPLE ONLINE AND REALTIME TRACKING, 2016)

SORT was one of the first tracking-by-detection algorithms designed explicitly for real-time use. Its pipeline is deliberately minimal: a Kalman filter predicts each existing track's next position using a constant-velocity motion model, and the Hungarian algorithm matches these predictions to the current frame's detections using Intersection-over-Union (IoU) as the cost metric. There is no appearance model at all — an object is identified purely by where it is expected to be, not what it looks like.

This simplicity makes SORT extremely fast, but also highlights its main weakness: because it relies only on motion and spatial overlap, it loses track of an object quickly if that object is occluded for more than a frame or two, or if two objects cross paths — both produce ID switches. SORT also silently discards any detection below a fixed confidence threshold, which means partially-occluded or blurry objects are dropped rather than tracked through the difficult frame.

3. DEEPSORT (2017)

DeepSORT directly targets SORT's main weakness — identity switches under occlusion — by adding a learned appearance descriptor. A small CNN, trained on a person re-identification dataset, produces an embedding vector for each detected bounding box. Association then uses a cascade: candidates are first filtered by a Mahalanobis distance on the Kalman filter's motion prediction, and then matched using cosine distance between appearance embeddings, prioritizing tracks that have been seen most recently.

This appearance information lets DeepSORT re-identify an object after a brief full occlusion, something motion-only SORT cannot do. The cost is a second neural network that must run on every detected box, every frame — this adds meaningful latency, particularly as the number of tracked objects grows, since the appearance CNN's cost scales with detection count.

4. BYTETRACK (2021)

ByteTrack takes a different, simpler path to the same problem: instead of discarding low-confidence detections before tracking (as SORT and DeepSORT both do), it keeps them and uses them in a second, more lenient association pass — this strategy is called BYTE. High-confidence detections are matched to tracks first, exactly as in SORT. Detections the detector was less sure about — often partially occluded or blurry objects — are then matched against any tracks still unmatched, rather than being thrown away.

This recovers many of the identity losses that motion-only SORT suffers under occlusion, without needing DeepSORT's appearance CNN at all. Because ByteTrack adds no extra network, it keeps almost all of SORT's speed while substantially closing the accuracy gap to appearance-based methods — this combination is the main reason it became a widely-adopted default in recent tracking pipelines, including this project's.

5. BOT-SORT (2022)

BoT-SORT builds on ByteTrack's association strategy and adds two further refinements. First, camera motion compensation (CMC): it estimates and corrects for the camera's own movement between frames (using image registration), which matters when the camera itself pans or shakes — without this, a moving camera looks identical to every tracked object moving, confusing the motion model. Second, an improved Kalman filter state that tracks bounding box aspect ratio more directly, giving more stable box predictions. BoT-SORT also supports an optional ReID (appearance) module, similar in spirit to DeepSORT's, for scenes with heavy occlusion.

The result is generally the most accurate of the four, particularly on moving-camera footage, at a moderate speed cost over ByteTrack — this project's own Assignment 2 experiments (Section 6) measured BoT-SORT at roughly 9–40% lower FPS than ByteTrack depending on model size, consistent with this added computation.

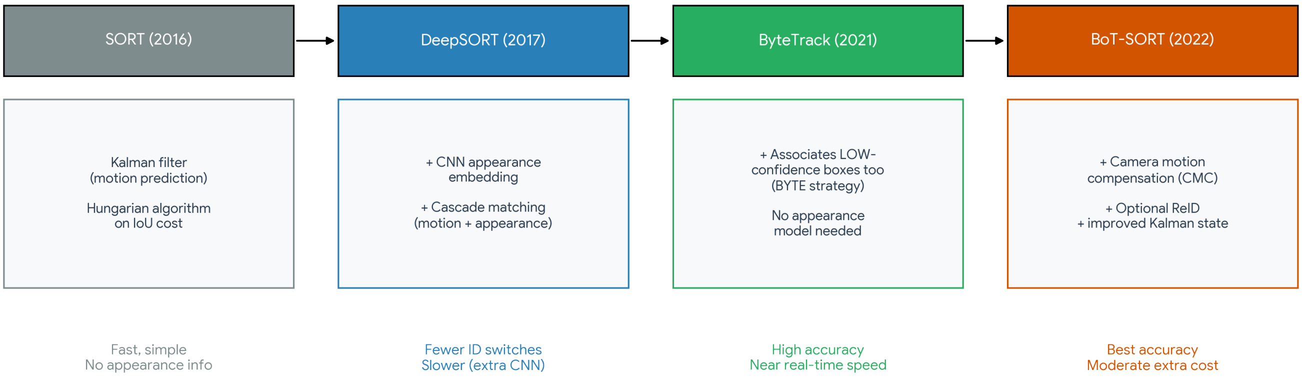

Figure 2 — What each algorithm added over its predecessor, and the resulting tradeoff.

6. COMPARISON: ACCURACY, SPEED, APPLICATIONS

The qualitative ranking below reflects widely-reported results on standard MOT benchmarks (e.g. MOT17/MOT20) in the original papers and subsequent literature. The speed figures for ByteTrack and BoT-SORT are cross-checked against this project's own Assignment 2 measurements (YOLOv8s, fp16, NVIDIA Quadro T1000) for grounding, rather than relying on literature alone.

ALGORITHM	RELATIVE ACCURACY	RELATIVE SPEED	TYPICAL APPLICATIONS
SORT	Lowest — frequent ID switches under occlusion	Fastest — no appearance model	High-throughput scenarios where brief ID errors are tolerable, e.g. basic people counting
DeepSORT	Better than SORT, especially through occlusion	Slower — extra CNN per detection	Surveillance / re-identification tasks where holding identity through occlusion matters more than raw speed
ByteTrack	High — recovers low-confidence detections	Near-SORT speed (no appearance model)	Real-time systems needing both speed and accuracy — the default choice for this project
BoT-SORT	Highest, esp. with camera motion	Moderate — CMC + optional ReID cost	Moving-camera footage (drones, PTZ cameras) and occlusion-heavy scenes such as crowded entrances

Empirical note from this project (Assignment 2): on identical input (YOLOv8s, confidence 0.3, GPU with fp16), ByteTrack ran at 12.97 FPS versus BoT-SORT's 11.74 FPS — a 9.5% gap — with identical detection counts and an identical ID-switch proxy count (71 for both) on a short test clip without heavy occlusion. This matches the expected pattern: BoT-SORT's extra accuracy machinery (CMC, optional ReID) shows its value specifically in occlusion or camera-motion scenarios, and provides little measurable benefit — only extra cost — on simpler footage where ByteTrack alone is already sufficient.

7. CONCLUSION

These four algorithms trace a clear line of development: SORT established the fast, motion-only baseline; DeepSORT showed that appearance information fixes SORT's occlusion weakness, at a real speed cost; ByteTrack showed that most of that same accuracy gain is available for free, just by not discarding low-confidence detections; and BoT-SORT layered camera motion compensation and optional re-identification on top of ByteTrack's association strategy for the highest accuracy, when the extra compute is affordable.

For this project's real-time video analytics platform, ByteTrack is the right default — it delivers most of the achievable accuracy at close to the lowest possible latency. BoT-SORT remains available as a switchable option (already implemented, see Assignment 1's app.py --tracker flag) for scenarios this project may later need to handle, such as a panning security camera or a crowded entrance with frequent occlusion, where its extra accuracy would justify its extra cost.

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